

CS 491 – Spring 2026
Software Engineering
General Course Information
(<https://github.com/paceuniversity/cs491>)
(<https://classes.pace.edu>)

“Learning without thinking is labour lost; thinking without learning is perilous”.
Confucius, *Analects*, II, 17.

Instructor: Dr. Christelle Scharff (she/her)
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Office Hours: MW 12-1 and 3:30-5:00, and On Zoom by appointment

General Information: CS 491 is a 4-credit course of the core Computer Science curriculum.

Prerequisites: Senior standing.

Course is taught using the web-based modality on Monday and Wednesday from 1:20 to 3:20. Classes will be face-to-face and synchronous online.

Course description: CS 491 introduces the concepts, tools and techniques of software engineering. Agile methodologies and Scrum are covered. It emphasizes the development of reliable and maintainable software via system requirements and specifications, software design methodologies including object-oriented design, implementation, integration and testing, software project management, life-cycle documentation, software maintenance and consideration of social or human factors issues. Students collaboratively define requirements, design, and implement a substantial project as a major part of the course.

Textbooks (recommended, not required):

- *Software Engineering*, Ian Sommerville, Addison Wesley, latest edition (<http://www.software-engin.com>).
- *The Mythical Man-Month*, Frederick P. Brooks Jr., Addison Wesley, 1995.
- *Software Requirements: Practical techniques for gathering and managing requirements throughout the product development cycle*, Karl E. Wiegers, 2nd edition, Microsoft Press, 2003.

Additional materials and recommended readings will be assigned in class and/or available from the web, including from the ACM and IEEE libraries.

CS 491 and BS in CS program outcomes:

CS 491 targets the following outcomes of the Bachelor of Science in Computer Science. The BS in CS is accredited by ABET.

- 2) Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program’s discipline.
- 3) Communicate effectively in a variety of professional contexts.
- 4) Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.
- 5) Function effectively as a member or leader of a team engaged in activities appropriate to the program’s discipline.
- 6) Apply computer science theory and software development fundamentals to produce computing-based solutions

CS 491 objectives and outcomes:

1. characterize the software development process models [2];
2. select an appropriate process model for a project with particular properties [6];
3. adapt general process models to the exigencies of the project [2, 6];
4. assign roles to software project team members and explain the responsibilities and interdependencies of each [5];
5. perform domain analysis through communications with application clients [3];

6. investigate the trade-offs of alternative designs and implementations relative to a project's requirements [2];
7. use the major UML diagrams to model both the static and dynamic aspects of a software [2];
8. explain the purpose, nature, and limitations of various forms of software [3];
9. apply different testing techniques to software [2];
10. discuss the major issues involved in project management [3, 5];
11. use project schedule estimation techniques [5];
12. use version control software (e.g. git) [2];
13. reflect on their own and their fellow teammates' contributions relative to: technical work, organizational contributions (e.g. planning, administration), writing, leadership, attitude, and initiative [3, 5];
14. reflect on the strengths that they used to the team's advantage [5];
15. reflect on the areas that need improvement relative to teamwork [5];
16. reflect on what had enhanced and what had handicapped their team's performance [5];
17. understand the ACM/IEEE Software Engineering Code of Ethics and Professional Practice [4].

Tools: Programming languages (JavaScript), framework (React), CASE tools (Discord, IDE, Git, GitHub, JUnit etc.), and databases technology (MySQL and FireBase etc.), AI APIs.

Topics:

- Introduction to software engineering
- Processes
- Agile and Scrum
- Requirements
- Design
- Testing
- Project management
- Tools to develop software and support software engineering
- New topics in software engineering (AI in Software Engineering)

The Following CS 491 Policies are Strictly Enforced

Course material: Course material (syllabus, guidelines, lecture notes, assignments, announcements, readings, etc...) will be available online. It is students' responsibility to consult the web (Classes and GitHub) regularly for new material.

Midterm/Final: There will be a midterm and a final. Readings and everything that is discussed as part of the class may be covered in the exams.

Project: There will be a software development project. The software to be developed will be a web application. A technology crash course will take place early in the semester. The goal of the project will be to develop an application to be available online and to build your portfolio. The software development process will mix Scrum and XP practices. The project will be realized in teams of 3-4 students. **As a senior-standing course, the project will require a consequent amount of independent work.** The project is guided with multiple deadlines and is documented in GitHub (only). The project is graded using rubrics, available in Classes and GitHub. All projects are encouraged to include an AI component. Not all students in a team get the same grade. The grades on the project are based on students' individual contributions. Contributions are monitored, including with GitHub.

Fairness: To ensure fairness, no work for the class is accepted after the last day of class.

Discord: Students will ask questions and collaborate with their team using Discord. They are encouraged to answer the questions of their classmates also. They must observe netiquette in the discussions.

Important Dates:

First day of class	Monday, January 26
No class (official Pace calendar)	Monday, February 16
Midterm	Monday, March 23
No class (official Pace calendar)	Monday, March 30 - Sunday, April 5
No class, Study day (official Pace calendar)	Wednesday, May 6
Final Project Presentation	Monday, May 11
Final Exam	Wednesday, May 13

Course Grades: All assignments related to the course have to be completed during the semester. No work will be accepted after the semester is over. Your overall course average for CS 491 will be computed using the following category weights:

Homeworks (including labs, quizzes etc.)	5%
Attendance and Session Assistantship	10%
Project	45%
Midterm	20%
Final exam	20%

Final letter grades are assigned in the following way:

A	90-100
A-	88-90
B+	85-88
B	82-85
B-	80-82
C+	75-80
C	70-75
C-	65-70
D+	60-65
D	55-60
F	0-55

A non-working software for the project will result in a grade of F. Deadlines are very important and have to be respected. Two deadlines that are not respected for the project will result in a reduction of one grade for the project. Students who are not presenting at the end of a spring automatically have a reduction of one grade for the project.

Attendance: Attendance is required. You have 1 hour to mark yourself present. Quickly will not be updated. It is your responsibility to mark yourself present on time. Over 3 absences will result in a reduction of one grade for the semester. Attendance is mandatory. Students who are not attending are penalizing their teams.

Comments: If you are not able to participate in the class activities and assignments, you have to notify the professor BEFORE the deadline.

Session assistant: Each student will be a session assistant at least once during the semester. The responsibilities of the session assistant include:

- Introduce each session with what was done during the last session and what are the next deadlines
- Check the access right for the documents that are posted after the session
- Take pictures/notes during the session
- Make a summary of what was done and what needs to be done for the next session

Netiquette: Our Pace community benefits from the richly unique experiences and individual diversity each of us bring. Intellectual growth and development happen when we engage in free and open discourse that challenges our own assumptions and beliefs. Together we all have the responsibility to create and maintain an environment where differences are respected and valued. To that end, we will challenge all manifestations of bias and discrimination to maintain a climate of mutual respect and civility. Whether you are learning in an online or on campus environment, the same expectations of courtesy and conduct apply. All classroom interactions should remain civil, respectful, and supportive. If you disagree with someone, aim to acknowledge your disagreement in a respectful way. Try responding with a question to open up further discussion (e.g., I'm not sure that I understand your point of view. Can you say more?). When working online, choose your words carefully. It's easy for someone to misinterpret your meaning when they can't see your expressions or hear the tone of your voice. Be careful when using sarcasm and humor. Without face-to-face communication, your comments may be misinterpreted.

Learning Commons: The [Learning Commons](#) uses an array of programs and a holistic approach to assist students with academic skills and content knowledge. We are dedicated to developing independent

learners through purposeful interactions with trained, well-qualified peer and professional staff. Services offered:

- Content Support Services including, content tutoring, exam review sessions, & content preparation/support workshops
- Academic Skills Services including small group peer mentoring, academic skills workshops, and individual academic development
- Writing Support Services including, writing tutoring & writing preparation/support workshops

Academic Integrity: Students in this course are required to adhere to [Pace University's Academic Integrity Code](#), which supports honesty and ethical conduct in the educational process. It educates students about what constitutes academic misconduct, helps to deter cheating and plagiarism, and provides a procedure for handling cases of academic misconduct. Students are expected to be familiar with the Code, which can be found in [Policies and Procedures](#). Individual schools and programs may have additional standards of academic integrity. Students are responsible for familiarizing themselves with the policies of the schools, programs, and courses in which they are enrolled.

Public Health Public health information and policies for the campus community are available on the Respiratory Virus (COVID-19, Flu, RSV) Guidance page. Pace recommends everyone continue to take steps to protect not only themselves but their colleagues, friends, and the campus by practicing good hand hygiene, staying home if you are sick, being up to date on vaccinations and boosters, and wearing a mask indoors when recommended by the CDC.

Pace University remains a mask-friendly community where mask use is always encouraged and supported. You may choose to wear a face mask, but they are not required outside of sick visits to healthcare settings.

Procedure for Students with Disabilities Who Wish to Obtain Reasonable Accommodations for a Course: The same rigorous standards for admission apply to students with and without a disability. In order to support the continued success of students with disabilities, the University prohibits discrimination on the basis of disability and is committed to ensuring equal access for students with disabilities to its facilities, programs, and activities. The University's commitment to equal educational opportunities for students with disabilities includes providing reasonable accommodations for the needs of students with disabilities.

To request an accommodation for a qualifying disability, a student must self-identify and register with the Disability Services Office for his or her campus. The Office of Disability Services is housed in the Counseling Center on the New York City and Pleasantville campuses. No one, including faculty, is authorized to evaluate the need and arrange for an accommodation except the Coordinator of Disability Services. Moreover, no one, including faculty, is authorized to contact the Coordinator of Disability Services on behalf of a student.

For further information, please see Information for [Students with Disabilities page](#).

Technological Resources:

- List of all [Pace Information Technology Services](#).
- For assistance with a technological concern contact the Pace Helpdesk at (914) 773-3648 or create a [help desk ticket](#).
- Visit the [Learning Remotely website](#)

Appropriate Use Policy for Information Technology: Respect for intellectual labor and creativity is vital to academic discourse and enterprise. This principle applies to work of all authors and publishers in all media. It encompasses respect for the right to acknowledgment, right to privacy and right to determine the form, manner and terms of publication and distribution.

Because electronic information is volatile and easily reproduced, respect for the work and personal expression of others is especially critical in computer environments. Violations of authorial integrity, including plagiarism, invasion of privacy, unauthorized access and trade secret and copyright violations, may be grounds for sanctions against members of the academic community.

Sex-Based Misconduct Policy and Procedure: Pace University is committed to providing a safe environment for every member of its community and to ensuring that no student, faculty or staff member is excluded from participation in or denied the benefits of any University program or activity on the basis of sex. Accordingly, the University prohibits the following forms of Sex-Based Misconduct: sexual assault, sexual harassment, gender-based harassment, dating violence, domestic violence, sexual exploitation and stalking.

Instructors are a non-confidential resource and have an obligation to report any information about sexual assault with the Executive Director of Institutional Equity and Title IX Coordinator – Bernard Dufresne, bdufresne@pace.edu, 163 Williams Street, Room 1017, (212) 346-1310. The Title IX/Affirmative Action

Office is responsible for investigating violations of the [sexual misconduct policy](#). Go here for more information about the Pace University sexual misconduct policy.

Members of the University community who believe that they have been subjected to Sex-Based Misconduct are encouraged to report such incidents to the University and, where applicable, to local law enforcement. Confidential resources include the University Counseling Centers, Offices of Sexual and Interpersonal Wellness and University Healthcare. Contact information for those offices may be found in the self-care section below.

Self Care: Your academic success in this course and throughout your college career depends heavily on your personal health and well-being. Stress is a common part of the college experience, and it often can be compounded by unexpected life changes outside the classroom. The Pace Community strongly encourages you to take care of yourself throughout the term, before the demands of midterms and finals reach their peak. Please feel free to talk with me about any difficulty you may be having that may impact your performance in this course as soon as it occurs and before it becomes unmanageable. Please know there are a number of other support services on campus that stand ready to assist you. I strongly encourage you to contact them when needed as well.

The Counseling Center Just In Case information supplies potentially life-saving mental health information to Pace University students, staff, and faculty, putting vital information and support options at your fingertips, just in case you or a friend needs help. Go to the [Just in Case](#) page on the Pace website or go to "Counseling Center" on the MyPace or the Pace Safe app.

During this academic year, the following information on [Coping Emotionally with COVID-19](#) may also be useful for you.